Adithya Shashikumaar Gurikar

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Education

University of Illinois Urbana-Champaign Expected May 2028
Master of Computer Science
Relevant Coursework: Database Systems, Advanced Data Management, Applied Machine Learning

Technical Skills

Programming Languages: Python, C, C++, Java, Javascript, SQL(MySQL), HTML/CSS
Backend & APIs: FastAPI, MongoDB, Redis, Git
Cloud & DevOps: AWS, Docker, Kubernetes
Libraries: LLMs, OpenAI API, Gemini API, Prompt Engineering, Multi-Agent Systems, SpaCy, PyTorch, TensorFlow, Scikit-learn, Pandas, SpaCy, ai4bharat-transliteration
Data: ETL/ELT Pipelines, Selenium, Beautiful Soup, Metabase, Tableau

Work Experience

Trainee Data Engineer Sep 2025 – Aug 2026
Eligarf Technologies Private Limited *Pune, India*

- Designed and shipped an LLM-powered astrology chatbot as a **FastAPI** service for iMeUsWe, a family heritage and astrology platform, delivering real-time personalized responses to thousands of users.
- Built **GenAI** based sentiment analysis and prompt optimization modules to improve astrologer-client interactions, increasing user engagement by **30%**.
- Designed and maintained **ETL/ELT** pipelines across MongoDB, Redis, and AWS for real-time ingestion, validation, and analytics, processing around **20,000** records per day.
- Partnered with product and testing teams to improve reliability, accuracy, and scalability.

Professional Services Consultant Intern Jul 2025 – Sep 2025
Aptos India Private Limited *Bengaluru, India*

- Configured, tested, and integrated data for CRM and Sales Audit systems for **5** major retail clients, executing test cases and documenting results in detailed reports.
- Resolved integration and usability issues during post-implementation support, ensuring stable client transitions.

Trainee Engineer Jan 2025 – Jul 2025
Eligarf Technologies Private Limited *Pune, India*

- Contributed to a **multi-agent GenAI** content-generation system using OpenAI and Gemini APIs to synthesize trend-based posts from Google Trends and iMeUsWe communities, generating over **250** posts per week.
- Applied **SpaCy NER models** (sm, lg, trf) to extract names and key entities from printed and handwritten PDFs and loading curated data into MongoDB, processing **100** documents a day.
- Built web scrapers with Selenium and Beautiful Soup to automate dynamic data extraction, processing **10,000+** records per day.

Projects

Automated Backup Service with Docker and Kubernetes | *Python, Docker, Kubernetes, Google Drive API, OAuth 2.0* GitHub

- Built a **containerized** Python service on the **Google Drive API** that authenticates with **OAuth 2.0** (automatic token refresh), creates a Drive folder, and uploads a directory's files using **resumable uploads**.
- Packaged the service with **Docker** and published the image to Docker Hub for portable, reproducible deployments across environments.
- Orchestrated scheduled backups on **Kubernetes** using a *CronJob* with a retry-on-failure policy, and **Persistent Volume Claims** to give the container access to the data being backed up.

Yet Another Distributed File System (YADFS) | *Python, Distributed Systems, Fault Tolerance* GitHub

- Engineered an **HDFS-inspired distributed file system** in Python where a NameNode manages the directory tree, file-to-block mappings, and per-DataNode capacity across **5 configurable DataNodes**.
- Implemented **64 KB block-level partitioning** with round-robin placement and a **replication factor of 3**, plus a CLI for **upload, read, list**, and **delete** operations (`put`, `cat`, `ls`, `mkdir`, `rm`, `tree`).
- Built **heartbeat checks** that validate DataNode logs against NameNode metadata to flag corrupted or unavailable nodes, and a **checkpointing** mechanism that restores NameNode metadata from a secondary copy after failure.

Spirulina Algae Growth Prediction and Yield Optimization | *Python, CatBoost, xGBoost, Scikit-learn, Hyperopt* GitHub

- Built a regression pipeline in a **4-member** team on NASA Ames raceway data (9 sensor features such as irradiance, pH, temperature, and nitrate) to predict Spirulina growth, benchmarking **6 models** including CatBoost, xGBoost, GBR, Stacking, SVR, and a neural network.
- Selected **CatBoost** (R^2 **0.988**, MSE **0.003**) as the best model, outperforming the neural network (R^2 0.917); used **Hyperopt** (Bayesian TPE) tuning, **k-fold cross-validation**, and interpolation/noise-injection augmentation to counter a small dataset and overfitting.
- Designed a **hybrid recommender system** that combines a content-based check of whether current raceway conditions support optimal growth with rule-based suggestions, serving the saved best model to help farmers adjust cultivation parameters.